

Section 6.5 — Non-Linear System

Math 103: College Algebra

Systems of Non-Linear Equations

Solve each system. Be sure to check for and discard extraneous solutions, if necessary.

1.
$$\begin{cases} 3x + 2y = 12 \\ xy = 6 \end{cases}$$

2.
$$\begin{cases} x^2 - y^2 = 8 \\ 2x - 5y = 1 \end{cases}$$

3.
$$\begin{cases} x^2 + y^2 = 41 \\ x^2 - y^2 = 9 \end{cases}$$

4.
$$\begin{cases} 2y^2 - x^2 = 28 \\ xy = -8 \end{cases}$$

5.
$$\begin{cases} 2x^2 + 5y^2 = 22 \\ x^2 + y^2 = 5 \end{cases}$$

6.
$$\begin{cases} 4x^2 - y^2 = 55 \\ 3x^2 - xy + y^2 = 45 \end{cases}$$

Note: this problem is a little more challenging.
Take your time and work carefully. Be sure to
check for extraneous solutions.